Customer Shopping Behavior Analysis

Project Overview

This project analyzes customer shopping behavior using a transactional dataset of 3,900 purchases. The primary objective is to identify key spending patterns, understand demographic influences on revenue, and evaluate the impact of loyalty programs (subscriptions) and discounts. The insights derived aim to provide actionable data-driven recommendations for marketing and inventory optimization.

Dataset Summary

The dataset provides a granular look at retail transactions with the following characteristics:

- Total Records: 3,900 entries.
- Key Features:
    - Demographics: Age, Gender, Location, and Subscription Status.
    - Transaction Details: Item Purchased, Category (Clothing, Accessories, Footwear, Outerwear), Purchase Amount (USD), and Season.
    - Behavioral Data: Review Rating, Shipping Type, Discount/Promo Code usage, and Previous Purchases.
- Data Quality: The raw data contained 37 missing values in the Review Rating column, which were addressed during the cleaning phase.

Data Preparation & Cleaning

Using the pandas library, the data underwent a rigorous cleaning and transformation process:

Handled missing values and standardized column names for easier SQL integration

```
df['Review Rating'] = df['Review Rating'].groupby(df['Category']).transform(lambda x: x.fillna(x.median()))
```

```
df.isnull().sum()
```

	0
Customer ID	0
Age	0
Gender	0
Item Purchased	0
Category	0
Purchase Amount (USD)	0
Location	0
Size	0
Color	0
Season	0
Review Rating	0
Subscription Status	0
Shipping Type	0
Discount Applied	0
Promo Code Used	0
Previous Purchases	0
Payment Method	0
Frequency of Purchases	0

dtype: int64

Age Segmentation: Created an age_group column to categorize customers into *Young Adult, Adult, Middle-aged,* and *Senior*.

```python
df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(' ', '_')
df = df.rename(columns={'purchase_amount_(usd)':'purchase_amount'})
df.columns
```

```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
       'purchase_amount', 'location', 'size', 'color', 'season',
       'review_rating', 'subscription_status', 'shipping_type',
       'discount_applied', 'promo_code_used', 'previous_purchases',
       'payment_method', 'frequency_of_purchases'],
      dtype='object')
```

```python
labels = ['Young Adult', 'Adult', 'Middle-aged', 'Senior']
df['age_group'] = pd.qcut(df['age'], q=4, labels=labels)

df[['age', 'age_group']].head(10)
```

	age	age_group
0	55	Middle-aged
1	19	Young Adult
2	50	Middle-aged
3	21	Young Adult
4	45	Middle-aged
5	46	Middle-aged
6	63	Senior
7	27	Young Adult
8	26	Young Adult
9	57	Middle-aged

Frequency Mapping: Converted categorical frequency_of_purchases into a numerical purchase_frequency_days to calculate engagement cycles.

```python
maps = {
    'Fortnightly' : 14,
    'Weekly': 7,
    'Monthly': 30,
    'Quarterly': 90,
    'Bi-Weekly': 14,
    'Annually': 365,
    'Every 3 Months': 90
}

df['purchase_frequency_days'] = df['frequency_of_purchases'].map(maps)
df[['frequency_of_purchases', 'purchase_frequency_days']].head(10)
```

	frequency_of_purchases	purchase_frequency_days
0	Fortnightly	14
1	Fortnightly	14
2	Weekly	7
3	Weekly	7
4	Annually	365
5	Weekly	7
6	Quarterly	90
7	Weekly	7
8	Annually	365
9	Quarterly	90

Exploratory Data Analysis

Key business questions were answered using SQL queries to extract deep-level insights:

Total Revenue generated male vs female customers

	gender	total_revenue
1	Male	157890
2	Female	75191

Customers used a discount but still spent more than the average purchase amount

	customer_id	purchase_amount
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88
12	29	94
13	32	79
14	33	67
15	35	91
16	37	69

Top 5 products with the highest average review rating

	item_purchased	Average Product Rating
1	Gloves	3.86142856223243
2	Sandals	3.84437500983477
3	Boots	3.81874999569522
4	Hat	3.80129870501432
5	Skirt	3.78481012809126

Comparison of average Purchase Amount between Standard and Express Shipping

	Average Purchase Amount	shipping_type
1	60	Express
2	58	Standard

Compare average spend and total revenue between subscribers and non-subscribers

	subscription_status	total_customer	average_revenue	total_revenue
1	Yes	1053	59.4919278252612	62645
2	No	2847	59.8651211801897	170436

5 products have the highest percentage of purchases with discounts applied

	item_purchased	discount_rate
1	Hat	50
2	Coat	49
3	Sneakers	49
4	Sweater	48
5	Pants	47

Segmentation of new, returning, and loyal customer

	customer_segment	total_customer
1	Returning	701
2	Loyal	3116
3	New	83

Top 3 most purchased products within each category

	category	item_purchased	total_orders	item_rank
1	Accessories	Jewelry	171	1
2	Accessories	Sunglasses	161	2
3	Accessories	Belt	161	3
4	Clothing	Blouse	171	1
5	Clothing	Pants	171	2
6	Clothing	Shirt	169	3
7	Footwear	Sandals	160	1
8	Footwear	Shoes	150	2
9	Footwear	Sneakers	145	3
10	Outerwear	Jacket	163	1
11	Outerwear	Coat	161	2

cutomers who are repeat buyers also likely to subsribe

	subscription_status	repeat_buyers
1	Yes	958
2	No	2518

Revenue contribution of each age group

Results | Messages

	age_group	revenue
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

Visualizations

The analysis culminated in an interactive dashboard (Customer Behaviour Dashboard.pbix) featuring:

- Executive KPIs: Total Revenue, Average Purchase Amount ($59.76), and Average Review Rating (3.75).
- Revenue by Age Group: Identified Young Adults as the top contributors with $62,143 in total revenue.
- Category Analysis: Visualized sales distribution showing *Clothing* and *Accessories* as the dominant categories.
- Subscription Breakdown: A donut chart highlighting that 27% of the customer base are active subscribers.

Business Recommendations

Based on the findings, the following strategies are recommended:

- Optimize Subscription Conversion: Since subscribers represent only 27% of the base, implement "Member-Only" discounts to convert the 73% non-subscribers.
- Targeted Marketing for Young Adults: Increase digital ad spend targeting the *Young Adult* demographic, as they are the highest revenue-generating group.
- Inventory Planning: Align stock levels for *Clothing* and *Accessories* categories with seasonal trends (Spring and Fall peaks) identified in the seasonal analysis.
- Loyalty Retention: Introduce a tiered rewards system for the "Returning" segment to accelerate their transition into "Loyal" customers.